Justin Ly

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Summary

Data Analyst with proven expertise in Python, SQL, Excel, and Power BI. Experienced in building ETL pipelines, statistical analysis, and interactive dashboards that drive data-driven decision making. Demonstrated success transforming complex datasets into actionable business intelligence through data modeling, proper data handling, predictive analytics, and data mining.

Work Experience

AI Red Team Intern, Berkeley Risk & Security Lab *(Berkeley, CA)* **June 2025-Present**

- Conducted statistical analysis of red teaming coverage gaps, identifying 40% underrepresentation in image/code attack vectors and delivering data-driven recommendations for future research allocation
- Constructed hypersonic missile simulation model with atmospheric and thermal effects allowing for up to 20,000 km trajectory predictions
- Developed interactive geospatial data visualization using Leaflet.js, enabling real-time mapping of flight paths and mission footprints with dynamic color coding and statistical overlays

Database Developer Intern, Morning Star *(Merced, CA)* **January 2025-May 2025**

- Engineered PostgreSQL database architecture for an automated seedling counting system, reducing manual counting time from ~10 minutes to under 10 seconds per 450-cell tray, improving operational efficiency by 95%
- Developed 15+ specialized database functions enabling statistical trend analysis, data retention policies, and anomaly reporting, increasing accuracy of plant health monitoring by ~94%
- Collaborated with cross-functional team to integrate DINOv2 and SVM machine learning outputs with database architecture

Data Engineer, University of California Merced *(Merced, CA)* **January 2024-May 2025**

- Tackled inefficient course registration affecting 9,000+ students by developing a data-driven recommendation system using SQL and Python frameworks, reducing advisor workload through automation, delivering the project within 6 months
- Addressed manual data processing bottlenecks by engineering an ETL pipeline with web scraping, data cleaning, and SQLite integration, improving registration efficiency by 97.83% through standardized data handling
- Resolved data structure limitations by designing a multi-relational database schema with normalized tables and SQL queries, delivering actionable insights while ensuring data integrity through bcrypt

Projects

Insurance Claims Communication Data Analytics Dashboard (Link) | *Power BI, Python, Pandas* **March 2025-May 2025**

- Transformed unstructured communication data into analyzable format through star schema data modeling and Python data cleaning techniques, enabling time-series analysis across 3+ years of claimant interactions
- Created interactive data visualizations with custom DAX measures to display business intelligence metrics and KPIs
- Implemented keyword pattern-based sentiment analysis, identifying that 65% of claimants were neutral despite 15.31-day average resolution times

Coffee Sales Dashboard (Link) | *Excel* **December 2024-January 2025**

- Refined chaotic sales data into strategic business intelligence insights through advanced Excel modeling (XLOOKUP, IF, INDEX-MATCH), enabling targeted inventory management
- Eliminated information overload by architecting a three-tier data visualization system using Excel's DAX measures, custom pivot tables with relational data modeling, and synchronized slicers, making sales correlations instantly accessible

Education

University of California Merced *Chancellor's List* **August 2021-May 2025**
Bachelor of Science, Computer Science & Engineering Major

Technical Skills

Programming Languages: SQL (SQLite, MySQL, PostgreSQL), Python, C++, JavaScript, R
Web Authoring, Frameworks, & Libraries: Pandas, HTML, CSS, Flask, Selenium, BeautifulSoup, PyTorch, NumPy
Technologies: Power BI, MS Office Suite, Firebase